curriculum vitæ of

Paul-Louis DELACOUR

Data scientist interested in theoretical research

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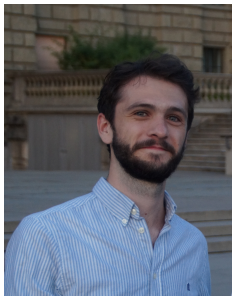

Education

Sep 2019 – Oct 2021 **MSc.** in Data Science. GPA : 5.25/6.00. ETH, Zürich , Switzerland

Relevant modules : Advanced algorithms, Optimization for Data Science, Probabilistic Artificial Intelligence.
Focus : Theoretical Computer Science and applied Machine Learning for Health Care.

Sep 2016 - July 2019 **BSc.** in Communication Systems. GPA : 5.35/6.00. EPFL, Lausanne, Switzerland

Relevant modules : Algorithms, Theory of Computation, Stochastic models in Communication.
Focus : Data Science and Theoretical Computer Science.

Sep 2013 – July 2016 **Baccalauréat** scientifique option Mathematics. Lycée du Grésivaudan, Meylan, France

Focus: Mathematics and Physics. Obtained with the Highest distinction.

Research Project

- **Master thesis** in the Theory of Combinatorial Algorithms group at ETH supervised by **Bernd Gärtner**.

  Worked on reduction of constrained convex programs to finding the sink in a unique sink orientation of hypercubes.

- **Bachelor thesis** in the THL4 algorithmic Lab at EPFL supervised by **Mikhail Kapralov**

  Worked on spectrally approximating large graphs with smaller graphs and its impact on clustering.

- **Data Science lab** in prediction of Psychiatric Disorders in a large Pediatric Sample.

  Predicted the severity of psychiatric disorders using EEG Data and efficiently represented those signals as disentangled factors to understand the nature of the information contained.

Applied Machine Learning in Health Care

- **ECG Heartbeat Classification : A Deep transferable Representation**.

  Classified heartbeat diseases, and inspected how transfer learning with multiple data sets could improve the results.

- **Prostate structure segmentation**

  Implemented a modified U-net architecture for segmentation of 3D prostate magnetic resonance images.

Skills

Learning background in `Optimization`, `Advanced Algorithms`, `Advanced Machine Learning` and `Reinforcement Learning`.

Strong knowledge of the programming languages : `Python` , `R` , `C` , `Java` , `Scala` with a focus on parallelism and concurent programming.

Spoken Languages

French : Native language.
English : Fluent speaker, Full Professional Proficiency.
Spanish : Limited Professional Proficiency.